# SOFT COMPUTING (CSC425) FEATURES SELECTION

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# FEATURES SELECTION

I. **Filter Methods**
II. Wrapper Methods
III. Embedded Methods

# FEATURES SELECTION

**Top reasons to use feature selection are:**

- It enables the machine learning algorithm to train faster.
- It reduces the complexity of a model and makes it easier to interpret.
- It improves the accuracy of a model if the right subset is chosen.
- It reduces overfitting.
- Avoids curse of dimensionality.

# FILTER METHODS

Filter methods are applied as pre-processing step.

The selection of features is **independent of any machine learning algorithms.**

Features are selected on the basis of their scores in various statistical tests for their correlation with the outcome variable.

These methods **are faster and less computationally** expensive than wrapper methods. When dealing with high-dimensional data, it is computationally cheaper to use filter methods.

Avoids overfitting but sometimes may fail to select best features.

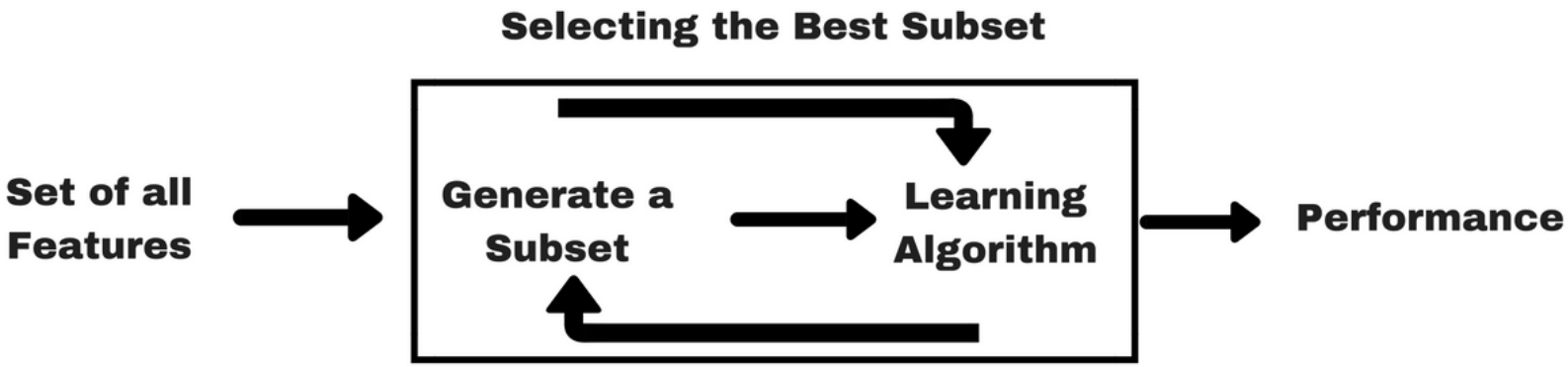
# FILTER METHODS

**Most known filter methods:**

- Information gain
- Chi square test
- Fisher score
- Correlation coefficient
- Variance Threshold
- MAD
- Dispersion ratio
- ANOVA

# WRAPPER METHODS

- Recursive feature elimination
- Backward elimination
- Forward selection
- Sequential feature selection algorithms
- Genetic algorithms



# WRAPPER METHOD

- In wrapper method, the feature selection algorithm exits as a **wrapper** around the predictive model algorithm and uses the same model to select best features.

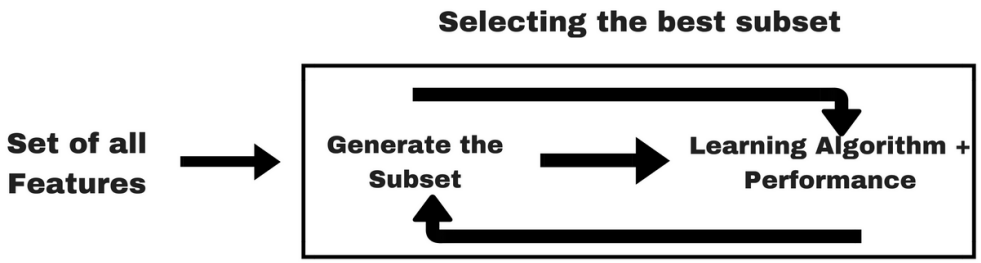
- Though computationally expensive and prone to overfitting, gives better performance.

# A WRAPPER METHOD EXAMPLE: SEQUENTIAL FEATURE SELECTION

- Sequential Forward Selection (SFS), a special case of sequential feature selection, is a greedy search algorithm that attempts to find the “optimal” feature subset by iteratively selecting features based on the classifier performance.
- We start with an empty feature subset and add one feature at the time in each round; this one feature is selected from the pool of all features that are not in our feature subset, and it is the feature that – when added – results in the best classifier performance.
- Since we have to train and cross-validate our model for each feature subset combination, this approach is much more expensive than a filter approach.

# EMBEDDED METHODS

- In **Embedded Methods**, the feature selection algorithm is integrated as part of the learning algorithm.
- It combines the qualities of  filter and wrapper methods.
- Some of the most popular examples of these methods are LASSO (L1 Regularization) and RIDGE regression (L2 Regularization), which have penalization functions to reduce overfitting.
- A learning algorithm takes advantage of its own variable selection process and performs feature selection and classification/regression at the same time.



# APPROACHES OF EMBEDDED METHODS

- **Lasso regression performs L1 regularization** which adds penalty equivalent to absolute value of the magnitude of coefficients.
- **Ridge regression performs L2 regularization** which adds penalty equivalent to square of the magnitude of coefficients.
- **Elastic Nets(L1 and L2)** balance between LASSO and RIDGE penalties.

# DIFFERENCE BETWEEN DIMENSIONALITY REDUCTION AND FEATURE SELECTION

- Feature selection selects a subset of the most significant features from a dataset, whereas dimensionality reduction transforms the data into a lower-dimensional space.

- Both strategies can be used to increase the performance and interpretability of a machine learning model, but they operate in distinct ways and serve different functions.

# DIMENSIONALITY REDUCTION VS. FEATURE SELECTION CON…

| | Dimensionality Reduction | Feature Selection |
|---|---|---|
| Definition | Reducing the number of dimensions in a dataset while retaining as much information as possible | Selecting a subset of the most relevant features from the dataset |
| Techniques | PCA, LDA, t-SNE, etc. | Filter methods, wrapper methods, embedded methods, etc. |
| Goal | Improve performance and interpretability of a machine learning model | Identify the most relevant and important features for predicting the target variable |
| Impact on data | Transforms the data into a lower-dimensional space | Keeps a subset of the original features |
| Impact on model | Can improve performance and interpretability by reducing complexity and risk of overfitting | Can improve performance and interpretability by reducing the number of features and risk of overfitting |

# METRICS FOR MODEL EVALUATION

The Confusion Matrix evaluates the ability of a classifier based on the following indicators

- TP (true positive): records correctly classified as Yes class
- FN (false negative): Incorrectly classified records as class No
- FP (false positive): Incorrectly classified records as class Yes
- TN (true negative) records correctly classified as class No

| | | Expected Class | |
|---|---|---|---|
| | | Class=Yes | Class=No |
| Actual Class | Class=Yes | TP | FN |
| | Class=No | FP | TN |

If the classification uses n classes, the confusion matrix will be $n \times n$

# ACCURACY

Accuracy is the most widely used metric to synthesize the information of a confusion matrix

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Equally, the frequency of the error could be used

$$\text{Error rate} = \frac{FP + FN}{TP + TN + FP + FN}$$

# ACCURACY LIMITIATIONS

Accuracy is not an appropriate metric if the classes contain a very different number of records. Consider a binary classification problem in which

- # Record of class 0 = 9990
- # Record of class 1 = 10

A model that always returns class 0 will have an accuracy of 9990/10000 = 99.9%

In the case of binary classification problems, the class "rare" is also called a positive class, while the class that includes most of the records is called a negative class

# PRECISION AND RECALL

Precision and Recall are two metrics used in applications where the correct classification of positive class records is more important

**Precision** measures the fraction of record results actually positive among all those who were classified as such

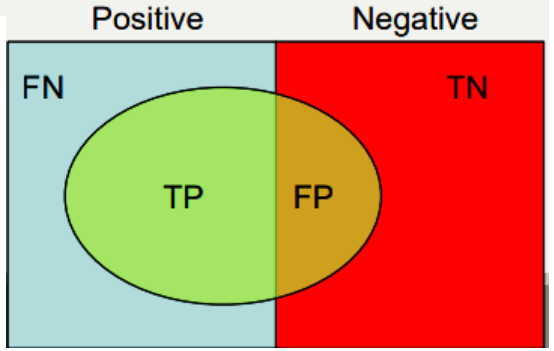
- High values indicate that few negative class records were incorrectly classified as positive.

**Recall** measures the fraction of positive records correctly classified

- High values indicate that few records of the positive class were incorrectly classified as negatives.

$$\text{Precision, } p = \frac{TP}{TP + FP}$$

$$\text{Recall, } r = \frac{TP}{TP + FN}$$



# PRECISION AND RECALL

Precision = 1 if all the positive records were actually detected

Recall = 1 if there are no false negatives

# EXAMPLE

- A model makes predictions and predicts 120 examples as belonging to the minority class, 90 of which are correct, and 30of which are incorrect.
- The precision for this model is calculated as:
- Precision=True Positives / (True Positives + False Positives)
- Precision = 90 / (90 + 30)
- Precision = 90 / 120
- Precision = 0.75

# F-MEASURE

A metric that summarizes precision and recall is called F-measure or F1-Score

This score takes false positives and false negatives into account.

$$\text{F-measure, F} = \frac{2rp}{r+p} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

F-measure represents the harmonic mean of precision and recall

If the F-measure is high, it means both precision and recall are.

F-measure is usually more useful than accuracy, especially if you have uneven class distribution.

Consider the confusion matrix given below for a binary classifier predicting the presence of a disease

The classifier made a total of 150 predictions

Out of those 150 cases, the classifier predicted "yes" 100 times, and "no" 50 times.

In reality, 100 patients in the sample have the disease, and 50 patients do not.

| | Predicted No | Predicted Yes |
|---|---|---|
| Actual No | TN = 45 | FP = 5 |
| Actual Yes | FN = 5 | TP = 95 |

$$Precision = \frac{TP}{Predicted\ Yes} = \frac{95}{100} = 95\%$$

$$Accuraccy = \frac{TN+TP}{TN+FP+FN+TP} = \frac{45+95}{150} = 93.33\%$$

**F1 score:** 66.67%

# CLASSIFICATION ERRORS

**Training error:** are mistakes that are made on the training set

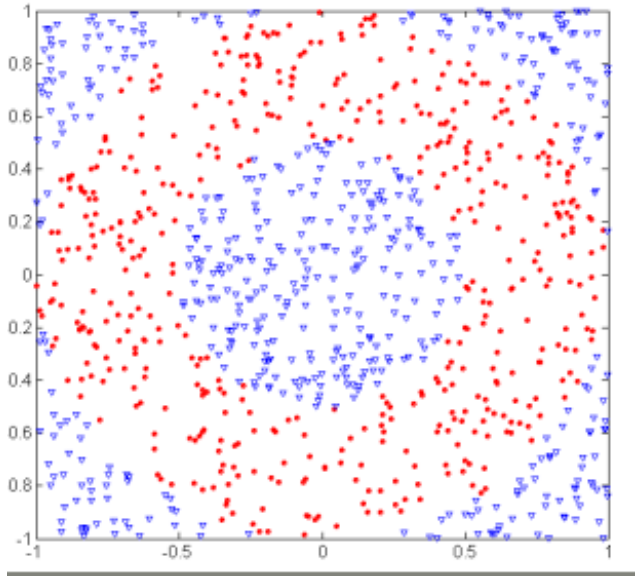
**Generalization error:** errors are made on the test set (i.e. records that have not been trained on the system).

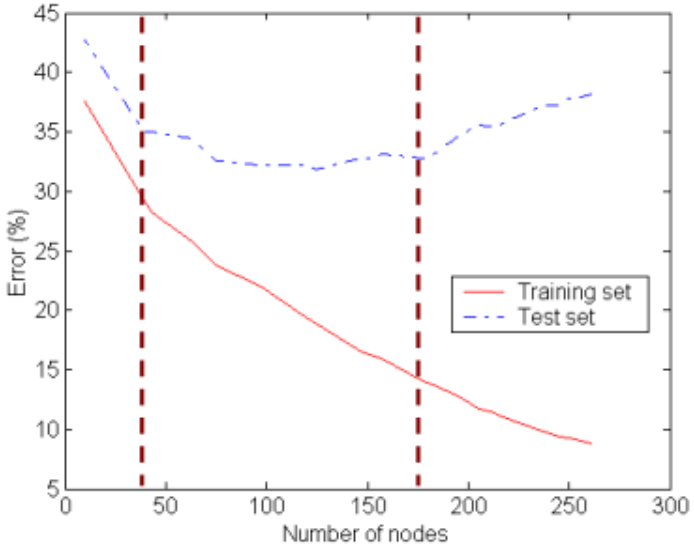
**Underfitting:** The model is too simple and does not allow a good classification or set training set or test set

**Overfitting:** The model is too complex, it allows a good classification of the training set, but a poor classification of the test set

- The model fails to generalize because it is based on the specific peculiarities of the training set that are not found in the test set (e.g. noise present in the training set)

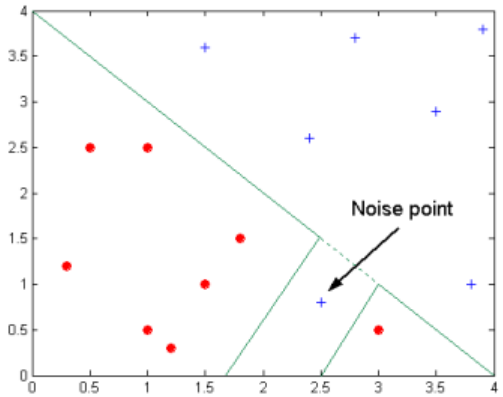
# UNDERFITTING AND OVERFITTING



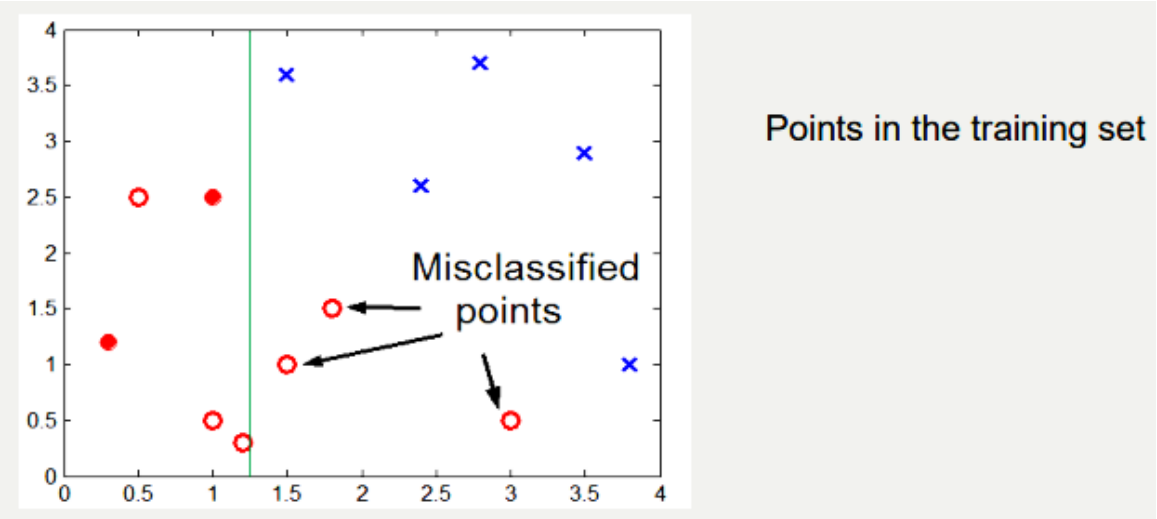



# OVERFITTING DUE TO NOISE

The boundaries of the areas are distorted due to noise.



# OVERFITTING DUE TO THE REDUCED SIZE OF THE TRAINING SET



# THANK YOU